DATE:

Module 5 Fourier Transforms.

Fourier Cosine Transform : $\hat{f_c}(\omega)$ (FCT)

[Read it as f c cap of ω]

$$\hat{f_c}(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cdot \cos\omega x \, dx$$

The inversion formula is given by

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty \hat{f_c}(\omega) \cos\omega x \, d\omega$$

Fourier sine Transform : $\hat{f_s}(\omega)$ (FST)

[Read it as f s cap of ω]

$$\hat{f_s}(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cdot \sin\omega x \, dx$$

The inversion formula is given by

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty \hat{f_s}(\omega) \cdot \sin\omega x \, d\omega$$

Other notations:

$\hat{f_c} = \mathcal{F}_c\{f(x)\}$, $\hat{f_s} = \mathcal{F}_s\{f(x)\}$

$\mathcal{F}_c^{-1}$ and $\mathcal{F}_s^{-1}$ for the inverse of $\mathcal{F}_c$ and $\mathcal{F}_s$ respectively.

Problems

1. Find the F.S.T & F.C.T of the fn

$$f(x) = \begin{cases} k & ; 0<x<a \\ 0 & ; x>a \end{cases}$$

ans:- $$\hat{f_c}(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos \omega x \, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{a} k \cos \omega x \, dx = \sqrt{\frac{2}{\pi}} k \left[\frac{\sin \omega x}{\omega}\right]_0^a$$

$$= \sqrt{\frac{2}{\pi}} \frac{k}{\omega} \left[\sin a\omega\right]$$

$$\hat{f_s}(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin \omega x \, dx.$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{a} k \sin \omega x \, dx = \sqrt{\frac{2}{\pi}} k \left[\cos \omega x\right]_0^a$$

$$= \sqrt{\frac{2}{\pi}} k \left[1 - \cos a\omega\right]$$

2. Find $\mathcal{F}_c(e^{-x})$ & $\mathcal{F}_s(e^{-x})$

$$\mathcal{F}_c(e^{-x}) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos \omega x \, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-x} \cos \omega x \, dx$$

$$= \sqrt{\frac{2}{\pi}} L\{\cos \omega x\} \text{ where } s=1$$

$$= \sqrt{\frac{2}{\pi}} \frac{s}{s^2+\omega^2} \text{ where } s=1$$

$$= \sqrt{\frac{2}{\pi}} \frac{1}{1+\omega^2}$$

$$\mathcal{F}_s(e^{-x}) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \sin wx \, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-x} \sin wx \, dx$$

$$= \sqrt{\frac{2}{\pi}} L\{\sin wx\} \quad \text{where } s=1$$

$$= \sqrt{\frac{2}{\pi}} \frac{w}{s^2+w^2} \quad " \quad = \sqrt{\frac{2}{\pi}} \frac{w}{1+w^2}$$

3. Find the F.C. Transform & F.S.T of e^{-ax}.

ans: $\mathcal{F}_c(w) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos wx \, dx$

$$= \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-ax} \cos wx \, dx.$$

$$= \sqrt{\frac{2}{\pi}} L\{\cos wx\} \quad \text{where } s=a$$

$$= \sqrt{\frac{2}{\pi}} \frac{s}{s^2+w^2} \quad = \sqrt{\frac{2}{\pi}} \frac{a}{a^2+w^2}$$

$$\mathcal{F}_s(w) = \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-ax} \sin wx \, dx$$

$$= \sqrt{\frac{2}{\pi}} L\{\sin wx\} \quad \text{where } s=a$$

$$= \sqrt{\frac{2}{\pi}} \frac{w}{a^2+w^2}$$

6. Find the cosine transform of $f(x) = \begin{cases} 1 & \text{if } 0<x<1 \\ -1 & \text{if } 1<x<2 \\ 0 & \text{if } x>2 \end{cases}$

$$\hat{f_c}(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos \omega x \, dx.$$

$$= \sqrt{\frac{2}{\pi}} \left\{ \int_0^1 \cos \omega x \, dx + \int_1^2 -1 \cos \omega x \, dx \right.$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{\sin \omega x}{\omega}\right]_0^1 - \left[\frac{\sin \omega x}{\omega}\right]_1^2$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{\sin \omega}{\omega} - \frac{\sin 2\omega}{\omega} + \frac{\sin \omega}{\omega}\right]$$

$$= \sqrt{\frac{2}{\pi}} \left\{\frac{2\sin \omega - \sin 2\omega}{\omega}\right\}$$

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7. Find $\hat{f_s}(\omega)$ for $f(x) = \begin{cases} x^2 & \text{if } 0<x<1 \\ 0 & \text{if } x>1 \end{cases}$

$$\hat{f_s}(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \sin \omega x \, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^1 x^2 \sin \omega x \, dx = \sqrt{\frac{2}{\pi}} \left[-\frac{x^2 \cos \omega x}{\omega} - \frac{2x \sin \omega x}{\omega^2} + \frac{2 \cos \omega x}{\omega^3} \right]_0^1$$

$$= \sqrt{\frac{2}{\pi}}\left[\frac{-\cos\omega}{\omega} + \frac{2\sin\omega}{\omega^2} + \frac{2\cos\omega}{\omega^3} - \frac{2}{\omega^3}\right]$$

8. Find the F.S.T of $e^{-|x|}$. Hence evaluate

$$\int_0^\infty \frac{\omega \sin\omega x}{1+\omega^2}\, d\omega .$$

$$\hat{f_s}(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-|x|} \sin\omega x \, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-x} \sin\omega x \, dx$$

$$= \sqrt{\frac{2}{\pi}}\, L\{\sin\omega x\} \text{ with } s=1$$

$$= \sqrt{\frac{2}{\pi}}\left[\frac{\omega}{s^2+\omega^2}\right] = \sqrt{\frac{2}{\pi}}\, \frac{\omega}{1+\omega^2}$$

Using inverse transform,

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty \hat{f_s}(\omega)\, \sin\omega x \, d\omega$$

$$= \sqrt{\frac{2}{\pi}} \int_0^\infty \sqrt{\frac{2}{\pi}}\, \frac{\omega}{1+\omega^2} \sin\omega x \, d\omega$$

$$= \frac{2}{\pi} \int_0^\infty \frac{\omega \sin\omega x}{1+\omega^2}\, d\omega .$$

$$\Rightarrow \int_0^\infty \frac{w \sin wx}{1+w^2}\, dw = \frac{\pi}{2} f(x) = \frac{\pi}{2} e^{-x}.$$

9. Find the F.C.T of $e^{-|x|}$ and deduce that

H.W

$$\int_0^\infty \frac{\cos wx}{1+w^2}\, dw = \frac{\pi}{2} e^{-x}$$

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Fourier Transform ($f^{\wedge}(\omega)$)

$$f^{\wedge}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)\, e^{-i\omega x}\, dx$$

The inverse of Fourier Transform is $f(x)$

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(\omega)\, e^{i\omega x}\, d\omega.$$

Note:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$e^{-i\theta} = \cos\theta - i\sin\theta$$

$$\frac{e^{i\theta} + e^{-i\theta}}{2} = \cos\theta$$

$$\frac{e^{i\theta} - e^{-i\theta}}{2i} = \sin\theta.$$

1) Find the fourier Transform of $f(x)=\begin{cases} 1 & \text{if } |x|<1 \\ 0 & \text{otherwise} \end{cases}$

$$\hat{f}(\omega)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(x)\cdot e^{-i\omega x}\,dx \qquad \text{Note } |x|<1 \Rightarrow -1<x<1$$

$$=\frac{1}{\sqrt{2\pi}}\int_{-1}^{1} 1\cdot e^{-i\omega x}\,dx$$

$$=\frac{1}{\sqrt{2\pi}}\left(\frac{e^{-i\omega x}}{-i\omega}\right)_{x=-1}^{x=1}$$

$$=\frac{1}{\sqrt{2\pi}(-i\omega)}\left(e^{-i\omega}-e^{i\omega}\right)$$

$$=\frac{1}{\sqrt{2\pi}(-i\omega)}\left(\cancel{\cos\omega}-i\sin\omega-(\cancel{\cos\omega}+i\sin\omega)\right)$$

$$=\frac{1}{\sqrt{2\pi}(-i\omega)}\left(-2i\sin\omega\right)$$

$$=\frac{2}{\sqrt{2\pi}}\,\frac{\sin\omega}{\omega}=\sqrt{\frac{2}{\pi}}\,\frac{\sin\omega}{\omega}$$

2. Find the fourier transform of

$$f(x) = \begin{cases} x & 0<x<a \\ 0 & \text{o.w} \end{cases}$$

$$f(\omega) = \frac{1}{\sqrt{2\pi}} \int_0^a \overset{I}{(x)} \overset{II}{(e^{-i\omega x})} \, dx$$ (integration by chain rule)

$$= \frac{1}{\sqrt{2\pi}} \left[(x)\left(\frac{e^{-i\omega x}}{-i\omega}\right) - (1)\left(\frac{e^{-i\omega x}}{(-i\omega)^2}\right) \right]_0^a$$

$$= \frac{1}{\sqrt{2\pi}} \left[\left(a \cdot \frac{e^{-i\omega a}}{-i\omega} - \frac{e^{-i\omega a}}{-\omega^2}\right) - \left(0 - \frac{e^0}{-\omega^2}\right) \right]$$

$$= \frac{1}{\sqrt{2\pi}} \left(\frac{a e^{-ia\omega}}{-i\omega} + \frac{e^{-ia\omega}}{\omega^2} - \frac{1}{\omega^2} \right)$$

3. $$f(x) = \begin{cases} e^{x}; & \text{if } -a<x<a \\ 0; & \text{otherwise} \end{cases}$$

$$f(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-a}^{a} e^{x} \cdot e^{-i\omega x} \, dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-a}^{a} e^{(1-i\omega)x} \, dx = \frac{1}{\sqrt{2\pi}} \left[\frac{e^{(1-i\omega)x}}{(1-i\omega)} \right]_{-a}^{a}$$

$$= \frac{1}{\sqrt{2\pi}(1-i\omega)} \left[e^{(1-i\omega)a} - e^{-(1-i\omega)a} \right]$$

$$= \frac{1}{\sqrt{2\pi}(1-i\omega)} \left[e^{a} \cdot e^{-ia\omega} - e^{-a} \cdot e^{ia\omega} \right]$$

4) Find the Fourier transform of $f(x)$ where $f(x) = \begin{cases} 1-x^2 ; & |x|<1 \\ 0 ; & |x|>1 \end{cases}$

Also show that $\int_0^\infty \cos\left(\frac{\omega}{2}\right)\left(\frac{\sin\omega - \omega\cos\omega}{\omega^3}\right) d\omega = \frac{3\pi}{16}$

Ans:- $\mathcal{F}\{f(x)\} = \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} (1-x^2)\, e^{-i\omega x}\, dx$

$$= \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} \underbrace{(1-x^2)}_{\text{even}} (\underbrace{\cos\omega x}_{\text{even}} - i\underbrace{\sin\omega x}_{\text{odd}})\, dx$$

$$= \frac{2}{\sqrt{2\pi}} \int_0^1 (1-x^2)\cos\omega x \, dx$$

$$= \sqrt{\frac{2}{\pi}} \left\{ (1-x^2)\left(\frac{\sin\omega x}{\omega}\right) - (-2x)\left(\frac{-\cos\omega x}{\omega^2}\right) + (-2)\cdot\left(\frac{-\sin\omega x}{\omega^3}\right) \right\}_{x=0}^{x=1}$$

...

$$= \sqrt{\frac{2}{\pi}} \left\{ \frac{-2\cos\omega}{\omega^2} + \frac{2\sin\omega}{\omega^3} \right\}$$

$$\hat{f}(\omega) = 2\sqrt{\frac{2}{\pi}} \left\{ \frac{\sin\omega - \omega\cos\omega}{\omega^3} \right\}$$

Taking inverse fourier transform, we have

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(\omega)\, e^{i\omega x}\, d\omega$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} 2\sqrt{\frac{2}{\pi}} \left(\frac{\sin\omega - \omega\cos\omega}{\omega^3} \right) e^{i\omega x}\, d\omega$$

$$= \frac{2}{\pi} \int_{-\infty}^{\infty} \left(\frac{\overset{\text{odd}}{\sin\omega} - \overset{\text{odd}}{\omega\cos\omega}}{\underset{\text{odd}}{\omega^3}} \right) \left(\overset{\text{even}}{\cos\omega x} + i\,\overset{\text{odd}}{\sin\omega x} \right)$$

$$= \frac{2}{\pi} \int_{-\infty}^{\infty} \left(\frac{\sin\omega - \omega\cos\omega}{\omega^3} \right) \cos\omega x\, d\omega$$

$$= \frac{2}{\pi} \cdot 2 \int_{0}^{\infty} \left(\frac{\sin\omega - \omega\cos\omega}{\omega^3} \right) \cos\omega x\, d\omega$$

$$\frac{4}{\pi}\int_0^\infty\left(\frac{\sin\omega-\omega\cos\omega}{\omega^3}\right)\cos\omega x\,d\omega=\begin{cases}1-x^2; & |x|<1\\ 0; & |x|>1\end{cases}$$

put $x=\frac{1}{2}$

$$\int_0^\infty\left(\frac{\sin\omega-\omega\cos\omega}{\omega^3}\right)\cos\frac{\omega}{2}\,d\omega=\frac{\pi}{4}\;1-\left(\frac{1}{2}\right)^2$$

$$=\frac{\pi}{4}\times\frac{3}{4}=\frac{3\pi}{16}$$

5) Obtain the Fourier Sine & Cosine transform of $f(x)=\frac{e^{-ax}}{x}$

Ans:- $\hat{f_s}(\omega)=\sqrt{\frac{2}{\pi}}\int_0^\infty\frac{e^{-ax}}{x}\cdot\sin\omega x\,dx$

$$=\sqrt{\frac{2}{\pi}}\int_0^\infty e^{-ax}\cdot\frac{\sin\omega x}{x}\,dx$$

$$=\sqrt{\frac{2}{\pi}}\,L\left\{\frac{\sin\omega t}{t}\right\}\text{ with } s=a$$

$$=\sqrt{\frac{2}{\pi}}\int_s^\infty\frac{\omega}{s^2+\omega^2}\,ds.$$

$$= \sqrt{\frac{2}{\pi}} \left[\tan^{-1} \frac{s}{w} \right]_{s=s}^{s=\infty}$$

$$= \sqrt{\frac{2}{\pi}} \tan^{-1}\infty - \tan^{-1}\frac{s}{w}$$

$$= \frac{\pi}{2} - \tan^{-1}\frac{a}{w}$$

$$\hat{f_c}(w) = \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{e^{-ax}}{x} \cos wx \, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-ax} \frac{\cos wx}{x} \, dx$$

$$\frac{d}{dw} \hat{f_c}(w) = \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-ax} \frac{-\sin wx}{x}$$

$$= -\sqrt{\frac{2}{\pi}} \int_0^\infty e^{-ax} \cdot \sin wx \, dx$$

$$=$$

$$= -\sqrt{\frac{2}{\pi}} \frac{w}{a^2+w^2}$$

$$f_c(w) = -\sqrt{\frac{2}{\pi}} \int \frac{w}{a^2+w^2} \, dw$$

$$= -\sqrt{\frac{2}{\pi}} \left\{ \frac{\log(a^2+w^2)}{2} + C \right\}$$

6) Obtain the Fourier Sine & Cosine transform of $f(x) = \frac{1}{x}$.

$$f_s^{\wedge}(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cdot \sin\omega x \, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} \frac{1}{x} \sin\omega x \, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-0x} \cdot \frac{\sin\omega x}{x} \, dx.$$

$$= \sqrt{\frac{2}{\pi}} \; L\left\{\frac{\sin\omega t}{t}\right\} \quad \text{with } s=0$$

$$= \sqrt{\frac{2}{\pi}} \int_s^{\infty} \frac{\omega}{s^2+\omega^2} \, ds$$

$$= \sqrt{\frac{2}{\pi}} \left[\tan^{-1}\left(\frac{s}{\omega}\right)\right]_s^{\infty}$$

$$= \sqrt{\frac{2}{\pi}} \; \tan^{-1}\infty - \tan^{-1}0$$

$$= \sqrt{\frac{2}{\pi}} \left\{\frac{\pi}{2} - 0\right\}$$

$$= \sqrt{\frac{\pi}{2}}$$

Odd and even functions.

Even functions

If $f(-x) = f(x)$ then, $f(x)$ is even.

eg:- ① all constant functions.
② $\cos x$, $\sec x$
③ polynomial fns like $x^2, x^4, x^6, x^8, \ldots$

are even functions.

Odd functions.

If $f(-x) = -f(x)$ then, $f(x)$ is odd

eg:- ① $\sin x$, $\operatorname{cosec} x$.
② polynomial functions like $x, x^3, x^5, x^7, \ldots$
are ~~ev~~ odd functions.

Product of odd & even functions.

eg:- 1) $\underset{(+)}{\overset{(\text{even})}{\cos x}} \cdot \underset{(-)}{\overset{(\text{odd})}{\sin x}} = \underset{(-1)}{\text{odd}}$ function

2) $\underset{(-)}{\overset{\text{odd}}{x}} \cdot \underset{(-1)}{\overset{\text{odd}}{\sin x}} = \underset{(+)}{\text{even}}$ function.

3) $\underset{(+)}{\overset{\text{even}}{\cos x}} \cdot \underset{(+)}{x^{2\,\text{even}}} = \underset{(+)}{\text{even}}$ function.

Result

$$\int_{-a}^{a} f(x)\,dx = \begin{cases} 2\int_0^a f(x)\,dx \;; & \text{when } f(x) \text{ even} \\ 0 \;; & \text{when } f(x) \text{ odd.} \end{cases}$$

eg:- 1) $$\int_{-\pi}^{\pi} \underbrace{\overset{\text{odd}}{\sin x} \cdot \overset{\text{even}}{\cos x}}_{\text{odd}}\, dx = 0$$

2) $$\int_{-3}^{3} \underbrace{\overset{\text{even}}{(x^2+1)}}_{\text{even}} \overset{\text{even}}{(\cos x)}\, dx = 2\int_0^3 (x^2+1)\cos x\, dx.$$

3) $$\int_{-\pi/2}^{\pi/2} \underbrace{\overset{\text{even}}{(\sin^2 x)} \cdot \overset{\text{even}}{(\cos x)}}_{\text{even}}\, dx = 2\int_0^{\pi/2} \sin^2 x \cdot \cos x\, dx.$$

4) $$\int_{-1}^{1} \underbrace{\overset{\text{odd}}{(x)} \cdot \overset{\text{odd}}{(\sin x)}}_{\text{even}}\, dx = 2\int_0^1 x \cdot \sin x\, dx.$$

Properties of Fourier Transform

I **Linearity property.**

The fourier transform is a linear operation. ie for any functions $f(x)$ and $g(x)$ whose fourier Transform exist and any constants a & b, the Fourier transform of $af(x)+bg(x)$ exists and

$$\mathcal{F}\{af+bg\} = a\,\mathcal{F}\{f(x)\} + b\,\mathcal{F}\{g(x)\}$$

proof:-

$$\mathcal{F}\{af(x)+bg(x)\} = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-i\omega x}(af+bg)\,dx$$

$$= \frac{1}{\sqrt{2\pi}}\left\{\int_{-\infty}^{\infty} e^{-i\omega x} a f(x)\,dx + \int_{-\infty}^{\infty} e^{-i\omega x}\, b g(x)\,dx\right\}$$

$$= a \cdot \mathcal{F}\{f(x)\} + b\,\mathcal{F}\{g(x)\}$$

$$F\{f^n(x)\} = (-i\omega)^n F(\omega)$$

II: Fourier Transform of Derivative of f(x)

Let $f(x)$ be continuous on x & $f(x) \to 0$ as $|x| \to \infty$, then

$$\mathcal{F}\{f'(x)\} = -i\omega\, \mathcal{F}\{f(x)\}$$

proof:-

Not needed

$$\mathcal{F}\{f'(x)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \overset{(I)}{e^{-i\omega x}}\, \overset{(II)}{f'(x)}\, dx$$

$$= \frac{1}{\sqrt{2\pi}} \left\{ \left(e^{-i\omega x} f(x)\right)_{-\infty}^{\infty} - \int_{-\infty}^{\infty} e^{-i\omega x}(-i\omega) f(x)\, dx \right\}$$

By integration by parts

$$= \frac{1}{\sqrt{2\pi}}\, i\omega \int_{-\infty}^{\infty} e^{-i\omega x} f(x)\, dx$$

$$= -i\omega\, \mathcal{F}\{f(x)\}$$

Since $f(x) \to 0$ as $|x| \to \infty$, the desired result follows.

IIIly. $\mathcal{F}\{f''(x)\} = (i\omega)^2 \cdot \mathcal{F}\{f(x)\}$

$$\mathcal{F}\{f'''(x)\} = (i\omega)^3\, \mathcal{F}\{f(x)\}$$

$$\vdots$$

$$\mathcal{F}\{f^n(x)\} = (i\omega)^n \cdot \mathcal{F}\{f(x)\} \ldots$$

III Convolution Theorem (with out proof)

$$\mathcal{F}\{f*g\} = \sqrt{2\pi}\ \mathcal{F}\{f\}\ \mathcal{F}\{g\}$$

This formula will help us ~~in~~ in solving P.D.E

IV Multiplication by x^n

$$\mathcal{F}\{x^n . f(x)\} = (-i)^n \frac{d^n}{dw^n} . \mathcal{F}\{f(x)\}$$

VI Self Resiprocal property – A fn $f(x)$ is said to be self resiprocal if its F.T is obtained by just replacing x by w.

Std. Result

$$\int_{-\infty}^{\infty} e^{-t^2}\, dt = \Gamma\tfrac{1}{2} = \sqrt{\pi}$$

$$\int_{0}^{\infty} e^{-t^2}\, dt = \frac{\sqrt{\pi}}{2}$$

1) Find the fourier transform of e^{-x^2}. Hence find $\mathcal{F}\{x \cdot e^{-x^2}\}$

Ans:- $$\mathcal{F}\{e^{-x^2}\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-i\omega x} \cdot e^{-x^2} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x^2 + i\omega x)} dx$$

complete the square of $x^2 + i\omega x$

$$x^2 + i\omega x = x^2 + i\omega x + \left(\frac{i\omega}{2}\right)^2 - \left(\frac{i\omega}{2}\right)^2$$

$$= \left(x + \frac{i\omega}{2}\right)^2 + \frac{\omega^2}{4}$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left[\left(x + \frac{i\omega}{2}\right)^2 + \frac{\omega^2}{4}\right]} dx$$

$$= \frac{1}{\sqrt{2\pi}} e^{-\frac{\omega^2}{4}} \int_{-\infty}^{\infty} e^{-t^2} dt$$

put $t = x + \frac{i\omega}{2}$

$dt = dx$

$$= \frac{1}{\sqrt{2\pi}} e^{-\frac{\omega^2}{4}} \sqrt{\pi}$$

$$= \frac{1}{\sqrt{2}} e^{-\frac{\omega^2}{4}}$$

We have $\mathcal{F}\{f'(x)\} = i\omega\, \mathcal{F}\{f(x)\}$

Here take $f(x) = e^{-x^2}$

$$f'(x) = e^{-x^2} \cdot (-2x)$$

$$x e^{-x^2} = \frac{f'(x)}{-2}$$

$$\mathcal{F}\{x \cdot e^{-x^2}\} = \mathcal{F}\left\{\frac{f'(x)}{-2}\right\}$$ (Fourier transf of derivatives)

$$= \frac{-1}{2}(i\omega)\, \mathcal{F}\{e^{-x^2}\}$$

$$= +\frac{1}{2}\, i\omega \frac{1}{\sqrt{2}}\, e^{\frac{-\omega^2}{4}}$$

$$= \frac{+i\omega}{2\sqrt{2}}\, e^{\frac{-\omega^2}{4}}$$

Fourier integral

We know that any periodic function $f_L(x)$ of period $2L$ can be represented by a Fourier series

$$f_L(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi}{L}x + b_n \sin\frac{n\pi}{L}x$$

If we let $L \to \infty$, the above infinite series becomes an integral from 0 to ∞ which represents $f(x)$

$$f(x) = \frac{1}{\pi}\left\{\int_0^{\infty} \cos wx \int_{-\infty}^{\infty} f(v)\cos wv \, dv + \sin wx \int_{-\infty}^{\infty} f(v)\sin wv \, dv\right\} dw.$$

Fourier integral Theorem

If $f(x)$ is piecewise continuous in every finite interval and has a right hand derivative and a left hand derivative at every point and then $f(x)$ can be represented by a Fourier integral

$$\boxed{\begin{aligned} f(x) &= \int_0^{\infty} \left[A(\omega)\cos\omega x + B(\omega)\sin\omega x\right] d\omega \\ \text{where } A(\omega) &= \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\cos\omega v \, dv \\ B(\omega) &= \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\sin\omega v \, dv \end{aligned}}$$

At a point where $f(x)$ is discontinuous the value of the Fourier integral equals the average of the left and right hand limits of $f(x)$ at that point.

Fourier Cosine integral. (F.C.I)

$$f(x) = \int_0^\infty A(\omega)\cos\omega x \, d\omega$$

where $A(\omega) = \frac{2}{\pi}\int_0^\infty f(v)\cos\omega v \, dv$

Fourier Sine integral (F.S.I)

$$f(x) = \int_0^\infty B(\omega)\sin\omega x \, d\omega$$

where $B(\omega) = \frac{2}{\pi}\int_0^\infty f(v)\sin\omega v \, dv$

1) Find the fourier integral representation of the function

$$f(x) = \begin{cases} 1 & \text{if } |x|<1 \\ 0 & \text{if } |x|>1 \end{cases}$$

and hence evaluate $\int_0^\infty \frac{\sin\omega \cos\omega x}{\omega} d\omega$.

Ans:- Given that $f(x) = \begin{cases} 1 & \text{if } -1<x<1 \\ 0 & \text{if otherwise} \end{cases}$

∴ Fourier integral represention is

$$f(x) = \int_0^{\infty} [A(\omega)\cos\omega x + B(\omega)\sin\omega x]\, d\omega$$

where $A(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\cos\omega v\, dv$

$$= \frac{1}{\pi}\int_{-1}^{1} 1\cdot \underbrace{\cos\omega v}_{\text{even}}\, dv$$

$$= \frac{2}{\pi}\int_0^1 \cos\omega v\, dv = \frac{2}{\pi}\left(\frac{\sin\omega v}{\omega}\right)_{v=0}^{v=1}$$

$$= \frac{2}{\pi}\frac{\sin\omega}{\omega}$$

$$B(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\sin\omega v\, dv$$

$$= \frac{1}{\pi}\int_{-1}^{1} 1\cdot \underbrace{\sin\omega v}_{\text{odd}}\, dv = 0$$

∴ F.I representation is

$$f(x) = \int_0^{\infty} \frac{2}{\pi}\frac{\sin\omega}{\omega}\cos\omega x\, d\omega.$$

ie $\int_0^\infty \frac{\sin\omega \cdot \cos\omega x}{\omega} d\omega = \frac{\pi}{2} f(x)$

$$= \frac{\pi}{2}\begin{cases} 1 & \text{if } |x|<1 \\ 0 & \text{if } |x|>1 \end{cases}$$

$$= \begin{cases} \pi/2 & \text{if } |x|<1 \\ \frac{\frac{\pi}{2}+0}{2} & \text{if } |x|=1 \\ 0 & \text{if } |x|>1 \end{cases}$$

$\because |x|=1$ is a point of discontinuity, at these points, the value of integral is $\frac{\frac{\pi}{2}+0}{2} = \frac{\pi}{4}$

$$= \begin{cases} \frac{\pi}{2} & \text{if } |x|<1 \\ \pi/4 & \text{if } |x|=1 \\ 0 & \text{if } |x|>1 \end{cases}$$

2) Find the F.I representation of

$$f(x) = \begin{cases} 0 & x<0 \\ 1/2 & x=0 \\ e^{-x} & x>0 \end{cases}$$

Ans:- $f(x) = \int_0^\infty [A(\omega)\cos\omega x + B(\omega)\sin\omega x] d\omega$

$$A(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\cos\omega v \, dv$$

$$= \frac{1}{\pi}\int_{0}^{\infty} e^{-v} \cos\omega v \, dv$$ (Compare with Laplace Transforms)

$$= \frac{1}{\pi} L\{\cos\omega t\}$$

$L\{\cos\omega t\}$
$= \int_0^\infty e^{-st}\cos\omega t \, dt$

$$= \frac{1}{\pi} \cdot \frac{s}{s^2+\omega^2}$$

$\omega = \omega$
$t = v$
$s = 1$

$$= \frac{1}{\pi} \frac{1}{1+\omega^2}$$

$$B(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\sin\omega v \, dv$$

$$= \frac{1}{\pi}\int_{0}^{\infty} e^{-v}\sin\omega v \, dv$$

$$= \frac{1}{\pi} L\{\sin\omega t\}$$

$$= \frac{1}{\pi} \frac{\omega}{s^2+\omega^2}$$

$s=1$
$\omega=1$

$$= \frac{1}{\pi} \frac{\omega}{1+\omega^2}$$

$\therefore$ F.I is

$$f(x) = \frac{1}{\pi} \int_0^\infty \left(\frac{\cos \omega x + \omega \sin \omega x}{1+\omega^2} \right) d\omega.$$

3) Find the f.I representation of

H.W

$$f(x) = \begin{cases} 2 & \text{if } |x|<1 \\ 0 & \text{if } |x|>1 \end{cases}$$

Ans:- $A(\omega) = \frac{4 \sin \omega}{\pi \omega}$ $\quad B(\omega) = 0$

4) Represent $f(x)$ as a fourier Cosine integral where $f(x) = \begin{cases} 1 & \text{if } 0<x<1 \\ 0 & \text{if } x>1 \end{cases}$

Ans:- fourier cosine integral is

~~$f(x) = \frac{2}{\pi} \int_0 \cos \omega x \int$~~

$$f(x) = \int_0^\infty A(\omega) \cos \omega x \, d\omega$$

$$A(\omega) = \frac{2}{\pi} \int_0^\infty f(v) \cos \omega v \, dv$$

$$= \frac{2}{\pi}\int_0^1 1\cdot \cos\omega v \, dv$$

$$= \frac{2}{\pi}\left(\frac{\sin\omega v}{\omega}\right)_0^1$$

$$= \frac{2}{\pi}\ \frac{\sin\omega}{\omega}$$

$\therefore$ F.I is $f(x) = \frac{2}{\pi}\int_0^\infty \frac{\sin\omega}{\omega}\cos\omega x \, d\omega$

5) Find the F.C.I of $f(x) = \begin{cases} x^2 ; & 0<x<1 \\ 0 ; & x>1 \end{cases}$

Ans:- $f(x) = \int_0^\infty A(\omega)\cos\omega x \, d\omega$.

$$A(\omega) = \frac{2}{\pi}\int_0^\infty f(v)\cos\omega v \, dv$$

$$= \frac{2}{\pi}\int_0^1 v^2 \cos\omega v \, dv \qquad f(x) = x^2, \ f(v) = v^2.$$

$$= \frac{2}{\pi}\left[(v^2)\left(\frac{\sin\omega v}{\omega}\right) - (2v)\left(\frac{-\cos\omega v}{\omega^2}\right) + (2)\left(\frac{-\sin\omega v}{\omega^3}\right)\right]_0^1$$

$$= \frac{2}{\pi}\left[\frac{\sin\omega}{\omega} + 2\cdot\frac{\cos\omega}{\omega^2} - 2\cdot\frac{\sin\omega}{\omega^3}\right]$$

$$\therefore f(x) = \frac{2}{\pi}\int_0^\infty \left[\frac{\sin\omega}{\omega} + \frac{2\cos\omega}{\omega^2} - \frac{2\sin\omega}{\omega^3}\right]\cos\omega x \, d\omega.$$

6. $f(x) = \begin{cases} \sin x & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$ Find the F.C.I

Ans:- $f(x) = \int_0^\infty A(\omega)\cos\omega x \, d\omega,$

$$A(\omega) = \frac{2}{\pi}\int_0^\pi \sin v \cos\omega v \, dv$$

$$= \frac{2}{\pi}\int_0^\pi \frac{1}{2}\left(\sin(v+\omega v) + \sin(v-\omega v)\right) dv$$

$$= \frac{1}{\pi}\int_0^\pi \left[\sin(1+\omega)v + \sin(1-\omega)v\right] dv$$

$$= \frac{1}{\pi}\left[\frac{-\cos(1+\omega)v}{1+\omega} - \frac{\cos(1-\omega)v}{1-\omega}\right]_0^\pi$$

$$= \frac{1}{\pi}\left[\left(\frac{-\cos(1+\omega)\pi}{1+\omega} - \frac{\cos(1-\omega)\pi}{1-\omega}\right)\right.$$

$$\left.\left(+\frac{1}{1+\omega} + \frac{1}{1-\omega}\right)\right] \ldots$$

$$= \frac{1}{\pi}\left[\frac{-\cos\omega\pi}{1+\omega} - \frac{-\cos\omega\pi}{1-\omega} + \frac{2}{1-\omega^2}\right]$$

$$= \frac{1}{\pi}\left[\cos\omega\pi\left(\frac{1}{1+\omega} + \frac{1}{1-\omega}\right) + \frac{2}{1-\omega^2}\right]$$

$$= \frac{1}{\pi}\left[\cos\omega\pi\left(\frac{2}{1-\omega^2}\right) + \frac{2}{1-\omega^2}\right]$$

$$= \frac{2}{\pi(1-\omega^2)}\left[\cos\omega\pi + 1\right]$$

$\therefore$ F. I is

$$f(x) = \int_0^\infty \left[\frac{2}{\pi(1-\omega^2)}(\cos\omega\pi + 1)\cos\omega x\right] d\omega.$$

7) $f(x) = \begin{cases} e^{-x} & \text{if } 0<x<a \\ 0 & \text{if } x>a \end{cases}$ Find F.C.I

Ans $f(x) = \int_0^\infty A(\omega) \cos\omega x \, d\omega$

$$A(\omega) = \frac{2}{\pi} \int_0^\infty f(v) \cos\omega v \, dv$$

$$= \frac{2}{\pi} \int_0^a e^{-v} \cos\omega v \, dv$$

$$= \frac{2}{\pi} \left\{ \frac{e^{-v}}{(-1)^2+\omega^2} \left[-\cos\omega v + \omega \sin\omega v \right] \right\}_0^a$$

$$= \frac{2}{\pi} \left[\frac{e^{-a}}{1+\omega^2} \left(-\cos\omega a + \omega \sin a\omega \right) + \frac{1}{1+\omega^2} \right]$$

$$\therefore f(x) = \frac{2}{\pi} \int_0^\infty \left[\frac{1 - e^{-a}\cos a\omega + \omega e^{-a} \sin a\omega}{1+\omega^2} \right] \cos\omega x \, d\omega$$

Q: $f(x) = \begin{cases} \cos x & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$ F.S.I

Ans:- F.S.I is $f(x) = \int_0^\infty B(\omega) \sin \omega x \, d\omega$

$$B(\omega) = \frac{2}{\pi} \int_0^\infty f(v) \sin \omega v \, dv$$

$$= \frac{2}{\pi} \int_0^\pi \cos v \cdot \sin \omega v \, dv$$

$$= \frac{2}{\pi} \cdot \frac{1}{2} \int_0^\pi \left[\sin(1+\omega)v - \sin(1-\omega)v \right] dv$$

$$= \frac{1}{\pi} \left[\frac{-\cos(1+\omega)v}{1+\omega} + \frac{\cos(1-\omega)v}{1-\omega} \right]_0^\pi$$

$$= \frac{1}{\pi} \left[\frac{\cos \pi\omega}{1+\omega} + \frac{-\cos \pi\omega}{1-\omega} - \left(\frac{-1}{1+\omega} + \frac{1}{1-\omega} \right) \right]$$

$d\omega$

$$= \frac{1}{\pi} \left[\cos \pi\omega \left(\frac{1-\omega-1-\omega}{1-\omega^2} \right) - \left[\frac{-1+\omega+1+\omega}{1-\omega^2} \right] \right]$$

$$= \frac{1}{\pi(1-\omega^2)} \left[-2\omega \cos \pi\omega - 2\omega \right]$$

$$\therefore f(x) = \int_0^\infty \frac{-2\omega}{\pi(1-\omega^2)} (\cos \pi\omega + 1) \sin \omega x \, d\omega$$

12) S.T $\int_0^\infty \left(\frac{\sin \pi\omega \sin \omega x}{1-\omega^2}\right) d\omega$

$$= \begin{cases} \frac{\pi}{2} \sin x & 0<x<\pi \\ 0 & x>\pi \end{cases}$$

Ans- Since the integrand contains $\sin \omega x$, we can say that it is the F.S.I representation of RHS.

Let $f(x) = \text{RHS} = \begin{cases} \frac{\pi}{2} \sin x & 0<x<\pi \\ 0 & x>\pi \end{cases}$

F.S.I is $f(x) = \int_0^\infty B(\omega) \sin \omega x \, d\omega$.

$$B(\omega) = \frac{2}{\pi} \int_0^\infty f(v) \sin \omega v \, dv$$

$$= \frac{2}{\pi} \int_0^\pi \pi/2 \sin v \cdot \sin \omega v \, dv$$

$$= \int_0^{\pi} \frac{1}{2}\left[\cos(1-\omega)v - \cos(1+\omega)v\right] dv$$

$$= \frac{1}{2}\left[\frac{\sin(1-\omega)v}{1-\omega} - \frac{\sin(1+\omega)v}{1+\omega}\right]_0^{\pi}$$

$$= \frac{1}{2}\left[\frac{\sin\pi\omega}{1-\omega} + \frac{\sin\pi\omega}{1+\omega}\right]$$

$$= \frac{1}{2}\left[\frac{(1+\omega)\cancel{\sin\pi\omega} + (1-\omega)}{1-\omega^2}\right]\sin\pi\omega$$

$$= \frac{1}{2}\left[\frac{2.\sin\pi\omega}{1-\omega^2}\right] = \frac{\sin\pi\omega}{1-\omega^2}$$

$$\therefore f(x) = \int_0^{\infty} \frac{\sin\pi\omega.\sin\omega x}{1-\omega^2}\, d\omega$$

Hence $\displaystyle\int_0^{\infty} \frac{\sin\pi\omega.\sin\omega x}{1-\omega^2}\, d\omega = \begin{cases} \frac{\pi}{2}\sin x & 0<x<\pi \\ 0 & x>\pi \end{cases}$

Hence proved.

11. S.T $\int_0^\infty \left(\frac{\sin\omega - \omega\cos\omega}{\omega^2}\right) \sin\omega x \, d\omega = \begin{cases} \frac{\pi}{2}x & \text{if } 0<x<1 \\ \frac{\pi}{4} & \text{if } x=1 \\ 0 & \text{if } x>1 \end{cases}$

Ans:- Note:- Since the integrand contains $\sin\omega x$, we can say that it is the F.S.I representation of RHS.

Let $f(x) = \begin{cases} \frac{\pi}{2}x & \text{if } 0<x<1 \\ \frac{\pi}{4} & \text{if } x=1 \\ 0 & \text{if } x>1 \end{cases}$

F.S.I of $f(x)$ is

$$f(x) = \int_0^\infty B(\omega) \sin\omega x \, d\omega$$

$$B(\omega) = \frac{2}{\pi}\int_0^\infty f(v) \sin\omega v \, dv$$

$$= \frac{2}{\pi} \int_0^1 \frac{\pi}{2}\, v \sin \omega v \, dv$$

$$= \left[(v)\left(\frac{-\cos \omega v}{\omega}\right) - \left(\frac{-\sin \omega v}{\omega^2}\right) \right]_0^1$$

$$= \frac{-\cos \omega}{\omega} + \frac{\sin \omega}{\omega^2}$$

$$= \frac{\sin \omega - \omega \cos \omega}{\omega^2}$$

$$\therefore f(x) = \int_0^\infty \left(\frac{\sin \omega - \omega \cos \omega}{\omega^2}\right) \sin \omega x \, d\omega$$

$$\text{i.e. } \int_0^\infty \left(\frac{\sin \omega - \omega \cos \omega}{\omega^2}\right) \sin \omega x \, d\omega = \begin{cases} \frac{\pi}{2} x & 0<x<1 \\ \frac{\pi}{4} & x=1 \\ 0 & x>1 \end{cases}$$

Hence the proof.

13) Show that $\int_0^\infty \frac{\cos\omega x + \omega \sin\omega x}{1+\omega^2}\, d\omega$

$$= \begin{cases} 0 & \text{if } x<0 \\ \pi/2 & \text{if } x=0 \\ \pi e^{-x} & \text{if } x>0 \end{cases}$$

Ans:- Let $f(x) = \begin{cases} 0 & \text{if } x<0 \\ \pi/2 & \text{if } x=0 \\ \pi e^{-x} & \text{if } x>0 \end{cases}$

F.I of $f(x)$ is

$$f(x) = \int_0^\infty A(\omega)\cos\omega x + B(\omega)\sin\omega x \, d\omega$$

$$A(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\cos\omega v \, dv$$

$$= \frac{1}{\pi}\int_0^\infty \pi e^{-v}\cos\omega v \, dv$$

$$= L\{\cos\omega v\} \quad \text{where } s=1$$

$$= \frac{s}{s^2+\omega^2} \quad \text{where } s=1$$

$$= \frac{1}{1+\omega^2}$$

...

$$B(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \sin \omega v \, dv$$

$$= \frac{1}{\pi} \int_{0}^{\infty} \pi e^{-v} \sin \omega v \, dv$$

$$= L\{\sin \omega v\} \quad \text{where } s=1$$

$$= \frac{\omega}{s^2+\omega^2} \quad \text{where } s=1$$

$$= \frac{\omega}{1+\omega^2}$$

$$\therefore f(x) = \int_{0}^{\infty} \frac{1}{1+\omega} \cos \omega x + \frac{\omega}{1+\omega^2} \sin \omega x \, d\omega$$

$$= \int_{0}^{\infty} \frac{\cos \omega x + \omega \sin \omega x}{1+\omega^2} \, d\omega$$